TUTORIAL 6

Invertible Matrices and the Determinant

MOTIVATION AND GUIDING IDEA

Motivation. A basis turns the construction of a vector into the determination of unique coefficients. An invertible matrix guarantees that every input can be reconstructed from its output. Elementary row operations are reversible steps, so row reduction provides both a reconstruction algorithm and an algebraic explanation of why it works.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Develop methods for choosing and extending bases, recognize invertible matrices and calculate their inverses, and connect row reduction with products of elementary matrices. Check each solution through its meaning for a system of equations.

MOTIVATION AND GUIDING IDEA

From a basis to unique reconstruction. The columns of a square matrix form a basis if and only if every right-hand-side vector can be expressed uniquely using those columns. This is the meaning of invertibility and explains why an inverse solves a system for every right-hand side.

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6.1 Invertible Matrices

DEFINITION 6.1

A matrix $A \in M_n(\mathbb{F})$, with $n \geq 1$, is **invertible** if there is $B \in M_n(\mathbb{F})$ satisfying $AB = BA = I_n$.

The unique inverse B is denoted A^{-1}.

Basic Properties

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RECALL AND TOOLS

- If A is invertible, so is A^{-1}.
- If A, B are invertible, so is AB, and $(AB)^{-1} = B^{-1}A^{-1}$.
- If A is invertible, so is A^T, and $(A^T)^{-1} = (A^{-1})^T$.
- The identity is invertible, with $I^{-1} = I$.
- A square zero matrix of positive order is not invertible.

6.1.1 Exercises on the Algebra of Invertible Matrices

EXERCISE 6.1

Let A, B be square matrices, with A invertible, such that $A = B^2$ and $B = A^2$. Please prove that B is the inverse of A, that is, $B = A^{-1}$.

WORKED SOLUTION

We have

$$\begin{aligned} AB &= AA^2 = A^3, \\ A^3 &= A^3(AA^{-1}) = A^4A^{-1} \\ &= (A^2A^2)A^{-1} = B^2A^{-1} = AA^{-1} = I. \end{aligned}$$

Since $B = A^2$, also $BA = A^2A = A^3 = AB = I$. Therefore $AB = BA = I$, and $B = A^{-1}$.

6.1.2 An Algorithm for Finding the Inverse

PROPOSITION 6.1

A square matrix A is invertible if and only if it is row-equivalent to the identity matrix.

Algorithm

1. Form $(A \mid I)$ and row-reduce, aiming to put the left block into **reduced row-echelon form**.
2. Inspect the resulting left block:
   - If it is I, then A is invertible, and the right block is its inverse: $(I \mid A^{-1})$.
   - If it is not I, then A is not invertible.

EXERCISE 6.2

Please determine which matrices are invertible. For each invertible matrix, please find its inverse by row reduction and please verify the result.

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 6 \\ -2 & -4 \end{pmatrix}.$$

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EXERCISE 6.2 CONTINUED

WORKED SOLUTION

$$\left(\begin{array}{ccc|ccc} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right).$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 + 2R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & 1 & 2 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right).$$

$$A^{-1} = \begin{pmatrix} -2 & 1 & 2 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \qquad AA^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2+2 & 1 & 2-2 \\ 1-1 & 0 & 1 \end{pmatrix} = I_3.$$

In B, the second row is $-2/3$ times the first. The operation $R_2 \leftarrow R_2 + (2/3)R_1$ creates a zero row, so B is not invertible.

6.2 Elementary Matrices

DEFINITION 6.2

- Elementary matrices are square matrices representing the elementary row operations used in row reduction.
- Applying an elementary row operation to a matrix is equivalent to multiplying it **on the left** by the corresponding elementary matrix.
- Each elementary matrix is obtained by applying its row operation to the identity matrix.

Consequence. Matrices A, B are row-equivalent if and only if there is a finite sequence of elementary matrices such that $B = E_1 E_2 \cdots E_k A$.

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1. **Interchanging rows** i and j is represented by $E^{(1)}_{i,j}$:

$$E^{(1)}_{i,j} = \begin{pmatrix} 1 & & & & \\ & 0 & & 1 & \\ & & 1 & & \\ & 1 & & 0 & \\ & & & & 1 \end{pmatrix}.$$

   - This matrix is its own inverse: interchange the same two rows again.

2. **Multiplying row** i by a nonzero scalar λ is represented by $E^{(2)}_{i,\lambda}$:

$$E^{(2)}_{i,\lambda} = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & \lambda & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}.$$

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- Its inverse is $E^{(2)}_{i,\lambda^{-1}}$: multiply the same row by the reciprocal scalar.

3. **Adding λ times row j** to row i, with $i \neq j$, is represented by $E^{(3)}_{i,j,\lambda}$. The entry λ is in row i, column j:

$$E^{(3)}_{i,j,\lambda} = \begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & \lambda & \\ & & & \ddots & & \\ & & & & 1 & \\ & & & & & 1 \end{pmatrix}.$$

- Its inverse is $E^{(3)}_{i,j,-\lambda}$: add $-\lambda$ times row j to row i.

EXERCISE 6.3

Please use the preceding reduction to write $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}$ as a product of elementary matrices.

WORKED SOLUTION

The reduction operations are $R_1 \leftrightarrow R_2$, then $R_3 \leftarrow R_3 - R_2$, then $R_1 \leftarrow R_1 + 2R_3$. Their matrices are

$$E_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$E_3E_2E_1A = I_3, \qquad A = E_1^{-1}E_2^{-1}E_3^{-1}$$
$$= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The last two factors multiply to $\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$. The first factor swaps the first two rows and recovers A. Each factor is elementary.

EXERCISE 6.4

Let $A, B, C \in \mathbb{R}^{n\times n}$ be invertible, and let $\alpha \in \mathbb{R}$ with $\alpha \neq 0$. Define

$$D := \alpha A^3 B^T C^2 A.$$

1. Please prove that D is invertible.
2. Please find an explicit expression for D^{-1} in terms of A, B, C, α.

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EXERCISE 6.4 CONTINUED

WORKED SOLUTION

1. Each of $\alpha I_n, A, A^3, B^T, C^2$ is invertible:
   - $\alpha \neq 0$, so αI_n is invertible.
   - A is invertible by assumption, so A^3 is invertible.
   - Since B is invertible, B^T is invertible with $(B^T)^{-1} = (B^{-1})^T$.
   - Since C is invertible, C^2 is invertible with $(C^2)^{-1} = (C^{-1})^2$.

   A product of invertible matrices is invertible, so D is invertible.

2. Write the ordered product $D = \alpha A^3 B^T C^2 A$. The inverse of a product is the product of the inverses in reverse order:

$$(XYZ)^{-1} = Z^{-1}Y^{-1}X^{-1}.$$

   Thus

$$D^{-1} = A^{-1}(C^2)^{-1}(B^T)^{-1}(A^3)^{-1}(\alpha I_n)^{-1}.$$

   Compute each factor:

$$(C^2)^{-1} = (C^{-1})^2, \quad (B^T)^{-1} = (B^{-1})^T, \quad (A^3)^{-1} = (A^{-1})^3, \quad (\alpha I_n)^{-1} = \frac{1}{\alpha} I_n.$$

   Substitution gives

$$D^{-1} = A^{-1}(C^{-1})^2(B^{-1})^T(A^{-1})^3 \cdot \frac{1}{\alpha} I_n = \frac{1}{\alpha} A^{-1}(C^{-1})^2(B^{-1})^T(A^{-1})^3.$$

   This is the required expression for D^{-1}.

EXERCISE 6.5

Let

$$A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \in \mathbb{R}^{n \times n}.$$

Please prove that A is invertible if and only if every $\lambda_i \neq 0$, and please find an explicit formula for A^{-1}.

WORKED SOLUTION

If A is invertible, then every $\lambda_i \neq 0$. Suppose A is invertible and, for a contradiction, that

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EXERCISE 6.5 CONTINUED

$\lambda_k = 0$ for some index k. Consider the standard vector

$$\vec{e}_k = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

with 1 in position k and zero in every other coordinate. Then

$$A\vec{e}_k = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \lambda_k \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \vec{0},$$

since $\lambda_k = 0$. Thus $A\vec{e}_k = \vec{0}$ with $\vec{e}_k \neq \vec{0}$, a nontrivial solution of $Ax = 0$. If A were invertible, left multiplication by A^{-1} would give $e_k = A^{-1}0 = 0$, a contradiction. Therefore every $\lambda_i \neq 0$. **If every $\lambda_i \neq 0$, then A is invertible; we find its inverse.** Assume all these scalars are nonzero. Define

$$B := \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right) = \begin{pmatrix} \frac{1}{\lambda_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\lambda_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\lambda_n} \end{pmatrix}.$$

Calculate

$$\begin{aligned} AB &= \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right) \\ &= \operatorname{diag}\left(\lambda_1 \frac{1}{\lambda_1}, \ldots, \lambda_n \frac{1}{\lambda_n}\right) \\ &= \operatorname{diag}(1, \ldots, 1) = I_n. \end{aligned}$$

The same calculation in reverse order gives $BA = I_n$. Hence B is the inverse of A, so

$$A^{-1} = B = \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right).$$

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6.2.1 Equivalent Conditions

PROPOSITION 6.2

For a square matrix $A \in M_n(\mathbb{F})$, the following conditions are equivalent:

1. A is invertible.
2. A is row-equivalent to I_n.
3. $\operatorname{rank}(A) = n$.
4. For every column vector $\vec{b} \in \mathbb{F}^n$, the system $A\vec{x} = \vec{b}$ has a solution.
5. For every column vector $\vec{b} \in \mathbb{F}^n$, the system $A\vec{x} = \vec{b}$ has a unique solution $\vec{x} \in \mathbb{F}^n$.

6.2.2 The Connection with the Homogeneous System

In the preceding assertions, take $\vec{b} = \vec{0}$. There is then a unique solution, the trivial solution. Thus:

PROPOSITION 6.3

Another equivalent condition: $A \in M_n(\mathbb{R})$ is invertible if and only if its homogeneous system has only the trivial solution. In other words, the only solution of $A\vec{x} = \vec{0}$ is $\vec{x} = \vec{0}$.

EXERCISE 6.6

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}, \qquad b = (\alpha, 2\gamma, 3\phi)^T, \qquad \alpha, \gamma, \phi \in \mathbb{R}.$$

Please solve $Ax = b$ in terms of the parameters and please prove that the system is homogeneous if and only if its unique solution is zero.

WORKED SOLUTION

$$x = A^{-1}b = \begin{pmatrix} -2 & 1 & 2 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ 2\gamma \\ 3\phi \end{pmatrix} = \begin{pmatrix} -2\alpha + 2\gamma + 6\phi \\ \alpha \\ -\alpha + 3\phi \end{pmatrix}.$$

Indeed, $x_2 = \alpha$, $x_1 - 2x_3 = 2\gamma$, and $x_2 + x_3 = 3\phi$. The solution is unique because A is invertible. The system is homogeneous exactly when $b = 0$, equivalently $\alpha = \gamma = \phi = 0$, which gives $x = 0$. Conversely, if $x = 0$ is a solution, then $b = A0 = 0$.

EXERCISE 6.7

$$A = I_3 + N, \qquad N = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}.$$

Please compute N^2, N^3, please find A^{-1} without row reduction, and please solve $Ax = (3, 8, 3)^T$.

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EXERCISE 6.7 CONTINUED

WORKED SOLUTION

$$N^2 = \begin{pmatrix} 0 & 0 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = 0.$$

$$(I+N)(I-N+N^2) = I - N + N^2 + N - N^2 + N^3 = I,$$

$$(I-N+N^2)(I+N) = I + N - N - N^2 + N^2 + N^3 = I.$$

$$A^{-1} = I - N + N^2 = \begin{pmatrix} 1 & 1 & -3 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$x = A^{-1}\begin{pmatrix} 3 \\ 8 \\ 3 \end{pmatrix} = \begin{pmatrix} 3+8-3\cdot 3 \\ 8-3\cdot 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}.$$

The only nonzero entry of N^2 is $(-1)\cdot 3 = -3$. Verification gives $Ax = (2-(-1), -1+3\cdot 3, 3)^T = (3,8,3)^T$.

Additional practice

EXERCISE 6.8 CHALLENGE

Let $u, v \in \mathbb{R}^n$, $n \geq 1$, and define $A = I_n + uv^T$.

1. Please prove that A is invertible if and only if $1 + v^T u \neq 0$, and please find its inverse in this case without using a determinant.
2. Please apply the formula to $u = (1,-1,0)^T$, $v = (2,0,1)^T$, and please solve $Ax = (5,0,-1)^T$.

WORKED SOLUTION

Write $P = uv^T$ and $c = v^T u \in \mathbb{R}$. Associativity gives

$$P^2 = (uv^T)(uv^T) = u(v^T u)v^T = cP.$$

If $1 + c \neq 0$, define $B = I_n - P/(1+c)$. Then

$$AB = (I_n + P)\left(I_n - \frac{P}{1+c}\right) = I_n + \left(1 - \frac{1}{1+c} - \frac{c}{1+c}\right)P = I_n.$$

Multiplication in the opposite order gives exactly the same expansion, so $BA = I_n$. Therefore

$$A^{-1} = I_n - \frac{uv^T}{1+v^T u}.$$

If $1 + c = 0$, then $c = -1$, so $u \neq 0$, whereas

$$Au = u + u(v^T u) = (1+c)u = 0.$$

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EXERCISE 6.8 CHALLENGE CONTINUED

An inverse would imply $u = 0$, a contradiction. This proves both directions.
For the given example,

$$v^T u = 2, \quad A = \begin{pmatrix} 3 & 0 & 1 \\ -2 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ \frac{2}{3} & 1 & \frac{1}{3} \\ 0 & 0 & 1 \end{pmatrix}.$$

$$x = A^{-1} \begin{pmatrix} 5 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} (5+1)/3 \\ 10/3 - 1/3 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}.$$

Verification: $Ax = (3 \cdot 2 - 1, -2 \cdot 2 + 3 + 1, -1)^T = (5, 0, -1)^T$.

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EXERCISE 6.9 CHALLENGE

Let $A \in M_n(\mathbb{R})$ satisfy $A^m = I_n$ for an integer $m \geq 2$, and define

$$P = \frac{1}{m} \sum_{k=0}^{m-1} A^k.$$

Please prove that $AP = PA = P$ and $P^2 = P$. Please find the inverse of $I_n + P$.

WORKED SOLUTION

Multiplication by A shifts the powers, so

$$AP = \frac{1}{m}(A + A^2 + \cdots + A^m) = P.$$

Since powers of A commute with A, we also have $PA = P$. Induction gives $A^k P = P$ for every $k \geq 0$. Thus

$$P^2 = \frac{1}{m} \sum_{k=0}^{m-1} A^k P = P.$$

Consequently

$$(I_n + P)(I_n - P/2) = I_n + P/2 - P^2/2 = I_n.$$

The reverse product is also I_n, since both factors are polynomials in P. Therefore $(I_n+P)^{-1} = I_n - P/2$.

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EXERCISE 6.10 CHALLENGE

Let $A, B \in M_n(\mathbb{R})$ and define the block matrix

$$C = \begin{pmatrix} A & B \\ B & A \end{pmatrix}.$$

Please prove that C is invertible if and only if $A+B$ and $A-B$ are invertible. In that case, please find C^{-1} in terms of $(A+B)^{-1}$ and $(A-B)^{-1}$.

WORKED SOLUTION

Define

$$J = \begin{pmatrix} I_n & I_n \\ I_n & -I_n \end{pmatrix}.$$

Since $J^2 = 2I_{2n}$, we have $J^{-1} = J/2$. Block multiplication gives

$$CJ = \begin{pmatrix} A+B & A-B \\ A+B & -(A-B) \end{pmatrix},$$

and therefore

$$J^{-1}CJ = \begin{pmatrix} A+B & 0 \\ 0 & A-B \end{pmatrix}.$$

Multiplication on either side by invertible matrices preserves invertibility. A block diagonal matrix is invertible exactly when both its diagonal blocks are invertible. This proves the equivalence. Writing $X = (A+B)^{-1}$ and $Y = (A-B)^{-1}$, we obtain

$$\begin{aligned} C^{-1} &= J \begin{pmatrix} X & 0 \\ 0 & Y \end{pmatrix} J^{-1} \\ &= \frac{1}{2} \begin{pmatrix} X+Y & X-Y \\ X-Y & X+Y \end{pmatrix}. \end{aligned}$$

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6.3 Uniqueness of Reduced Row Echelon Form

PROPOSITION 6.4

Every $A \in M_{m\times n}(F)$ has a unique reduced row echelon form.

Proof. Gaussian elimination proves existence. Suppose R, S are reduced row echelon matrices row-equivalent to A. Every row operation and its inverse express new rows as linear combinations of old rows. Thus exactly the same vectors can be obtained as combinations of the rows of R and of S.

Let the nonzero rows of R be $r_1, \dots, r_t$, with pivot positions $p_1 < \cdots < p_t$. In a nonzero combination, if c_k is the first nonzero coefficient, then the first nonzero entry of $\sum_i c_i r_i$ is in position p_k and equals c_k. Therefore the possible first nonzero positions of all such combinations are exactly $p_1, \dots, p_t$. The same set is obtained from S, so both matrices have the same pivot positions and the same number of nonzero rows.

Write row i of S as $s_i = \sum_j c_j r_j$. Entry p_j gives $c_j = (s_i)_{p_j}$, since a pivot column of R has one 1 and zeros elsewhere. Since S is also reduced, $(s_i)_{p_j}$ is 1 if $i = j$ and zero otherwise. Hence $s_i = r_i$. The zero rows also agree, so $S = R$.

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EXERCISE 6.11

Let $A \in M_{m\times n}(F)$ and $B \in M_{k\times n}(F)$. Suppose $Ax = 0$ and $Bx = 0$ have the same solutions. Please prove that the nonzero rows of the reduced row echelon forms of A and B agree.
Hint. Please check when a solution can satisfy $x_j = 1$ and $x_{j+1} = \cdots = x_n = 0$.

WORKED SOLUTION

Let the reduced forms be R, S. Row operations preserve solutions, so $Rx = 0$ and $Sx = 0$ also have the same solution set.
In a reduced system, x_j is free if and only if there is a solution with $x_j = 1$ and all later variables zero. If j is free, set it to 1 and every other free variable to zero. Every later pivot variable is zero, since its equation involves only free variables to its right. If j is a pivot, its equation is

$$x_j + \sum_{\ell>j} r_{i\ell}x_\ell = 0,$$

and the requested conditions would give $1 = 0$.
The solution set therefore determines the free positions and hence the pivot positions. Now set one free variable $x_j = 1$ and all other free variables to zero. The equation with pivot p_i gives

$$x_{p_i} + r_{ij} = 0, \qquad r_{ij} = -x_{p_i}.$$

Thus every entry in a free column is determined by the solutions. The pivot columns are already fixed by reduced form. Hence the nonzero rows of R, S agree. If there are no free variables, both forms have the same rows of I_n, so the conclusion still holds.

EXERCISE 6.12

Let p be prime and let $m, n \geq 1$. Please find the number of $m \times n$ reduced row echelon matrices over $\mathbb{Z}_p$ with exactly one pivot.

WORKED SOLUTION

There is one nonzero row, which is the first row. If its pivot is in column j, the preceding entries are zero, the pivot is 1, and each of the $n-j$ later entries has p choices. This gives p^{n-j} matrices with a pivot in column j.
The pivot position is unique, so the cases are disjoint. Summing gives

$$\sum_{j=1}^{n} p^{n-j} = 1 + p + \cdots + p^{n-1} = \frac{p^n - 1}{p - 1}.$$

The division here is ordinary integer division for counting, not an operation in the field. For example, $p = 3, n = 3$ gives $9 + 3 + 1 = 13$ forms, for every $m \geq 1$.

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6.4 The Determinant over a Field

DEFINITION 6.3

Let F be a field and let $n \geq 1$. A determinant is a function $D : M_n(F) \to F$ satisfying:

1. Linearity in each row separately, with the other rows fixed.
2. $D(A) = 0$ whenever two rows of A are equal. This is called **alternation**.
3. $D(I_n) = 1$.

We shall prove existence and uniqueness, and then write det for this function.

NOTATION AND CONVENTIONS

A permutation of $1, \ldots, n$ is an ordering of these numbers without repetitions. Their set is denoted by S_n. For $\sigma \in S_n$, let $\mathrm{inv}(\sigma)$ count the pairs $i < j$ with $\sigma(i) > \sigma(j)$, and define

$$\mathrm{sgn}(\sigma) = (-1)^{\mathrm{inv}(\sigma)}.$$

Swapping two positions changes the sign. Indeed, an adjacent swap changes the parity of the inversion count, and a swap of nonadjacent positions is a composition of an odd number of adjacent swaps.

PROPOSITION 6.5

The determinant exists and is unique over every field.
Existence. For $A = (a_{ij})$, define

$$D(A) = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

Every product uses exactly one entry from each row, so the expression is linear in each row. If rows r, s are equal, pair each permutation with the one obtained by swapping $\sigma(r), \sigma(s)$. Their products agree and their signs are opposite, so each pair cancels. At I_n, only the identity permutation contributes, giving 1. All three properties hold.
Uniqueness. Suppose D has these properties. Let e_j^T be the row with 1 in position j and zeros elsewhere. Alternation and linearity first give

$$0 = D(\ldots, u+v, \ldots, u+v, \ldots) = D(\ldots, u, \ldots, v, \ldots) + D(\ldots, v, \ldots, u, \ldots).$$

The two terms with equal rows vanish. Thus swapping rows changes the sign, and hence

$$D(e_{\sigma(1)}^T, \ldots, e_{\sigma(n)}^T) = \mathrm{sgn}(\sigma).$$

Write each row as $\sum_j a_{ij} e_j^T$ and expand using linearity in all rows. Any term with two repeated choices vanishes. The remaining terms correspond to permutations, so necessarily

$$D(A) = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

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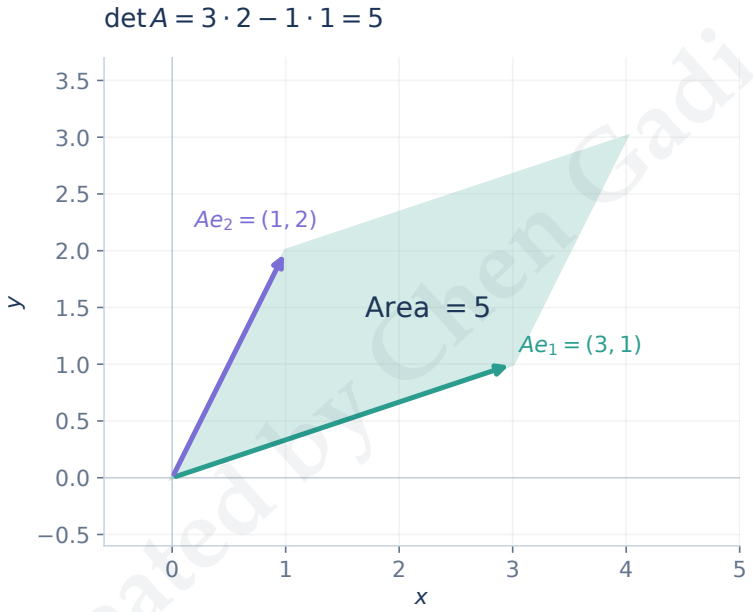
This is precisely the constructed function. The proof also works over $\mathbb{Z}_2$: cancelling opposite terms does not require division by 2.

NOTE

The geometric interpretation requires a real matrix. In $\mathbb{R}^2$, $|\det A|$ is the area scale factor; in $\mathbb{R}^3$, it is the volume scale factor. If $\det A = 0$, the image has smaller dimension. If $\det A \neq 0$, the sign indicates whether orientation is preserved or reversed.

6.5 The Determinant

The determinant is defined for square matrices. In the real plane, its absolute value is the area scale factor; its sign indicates whether orientation is preserved or reversed.



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Figure 6.1. A linear map sends the unit square to a parallelogram. The parallelogram's area is the absolute value of the determinant.

DEFINITION **6.4**

Let $n \in \mathbb{N}$. A function

$$\det : M_n(\mathbb{R}) \longrightarrow \mathbb{R}$$

is called the **determinant** if, for every $A \in M_n(\mathbb{R})$, it has the following properties:

1. **Interchanging rows (change of sign):** For $1 \leq i < j \leq n$, if $E^{(1)}_{i,j} \in M_n(\mathbb{R})$ is the elementary matrix interchanging rows i and j, then

$$\det\left(E^{(1)}_{i,j} A\right) = -\det(A).$$

2. **Multiplying a row by a scalar (homogeneity):** For $1 \leq i \leq n$ and $\lambda \in \mathbb{R} \setminus \{0\}$, if $E^{(2)}_{i,\lambda} \in M_n(\mathbb{R})$ is the elementary matrix multiplying row i by λ, then

$$\det\left(E^{(2)}_{i,\lambda} A\right) = \lambda \det(A).$$

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3. **Adding a multiple of another row (invariance):** For $1 \leq i \neq j \leq n$ and $\lambda \in \mathbb{R}$, if $E^{(3)}_{i,j,\lambda} \in M_n(\mathbb{R})$ is the elementary matrix performing $R_i \leftarrow R_i + \lambda R_j$, then

$$\det\left(E^{(3)}_{i,j,\lambda} A\right) = \det(A).$$

4. **Normalization:**

$$\det(I_n) = 1.$$

RECALL AND TOOLS

Two useful basic properties:

- If $A = (a_{ij}) \in M_n(\mathbb{R})$ is upper triangular, meaning that $a_{ij} = 0$ for every $i > j$, then

$$\det(A) = \prod_{k=1}^{n} a_{kk}.$$

- If rows i and j are equal for some $i \neq j$, or if A has a zero row, then $\det(A) = 0$. Similarly, two equal columns or a zero column imply $\det(A) = 0$.

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